MODELING VISUALIZATION AS AN AFFECTIVE TRIGGER FOR HUMAN DECISION-MAKING: THE VP-RAFIAH FRAMEWORK

AUTHORS

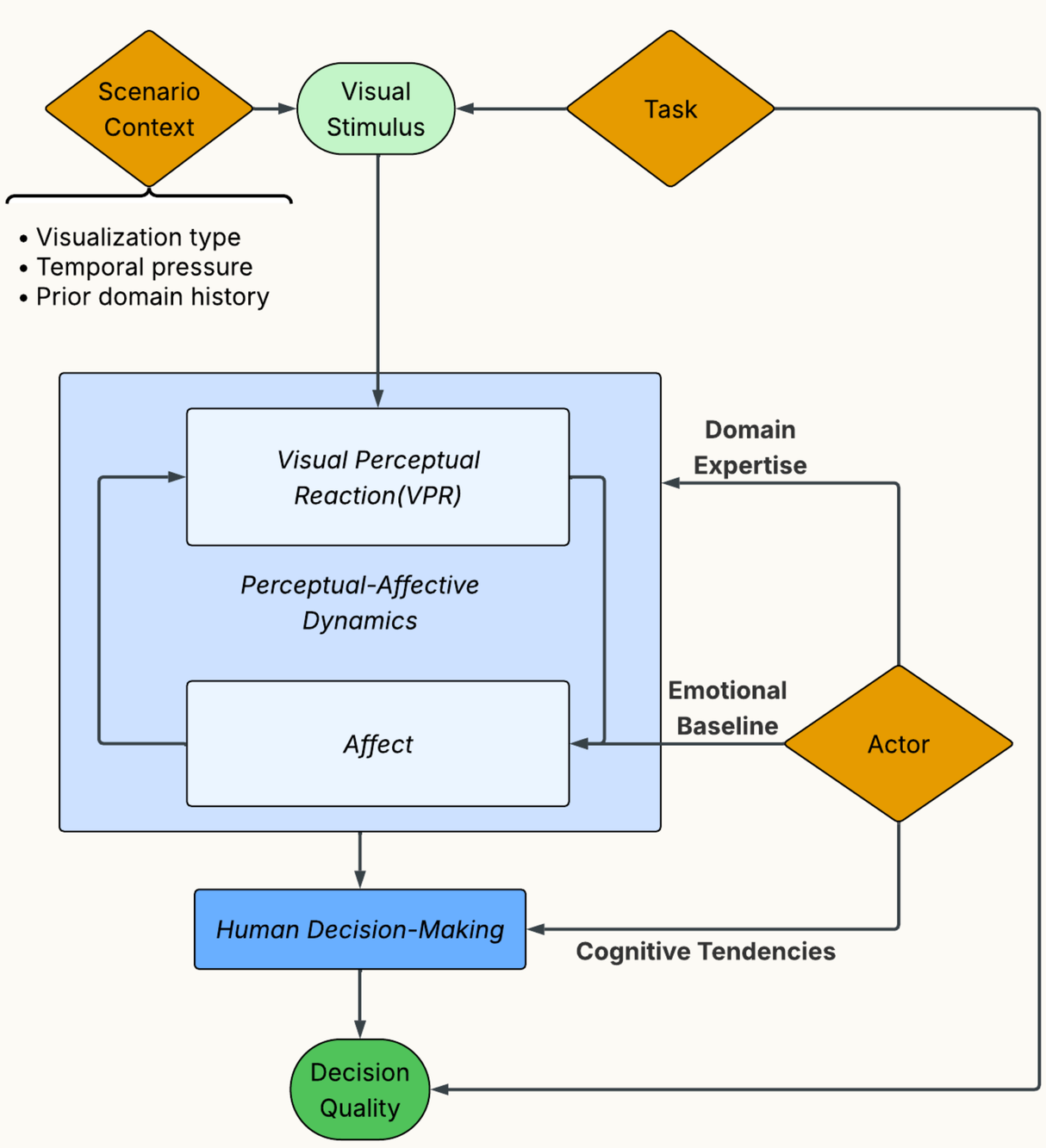
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INTRODUCTION

Visualizations are designed to support decisions, but there is no formal model for how seeing a visualization moves through a person's emotions and into decision quality. This work presents a novel framework at this intersection.

FRAMEWORK DESIGN



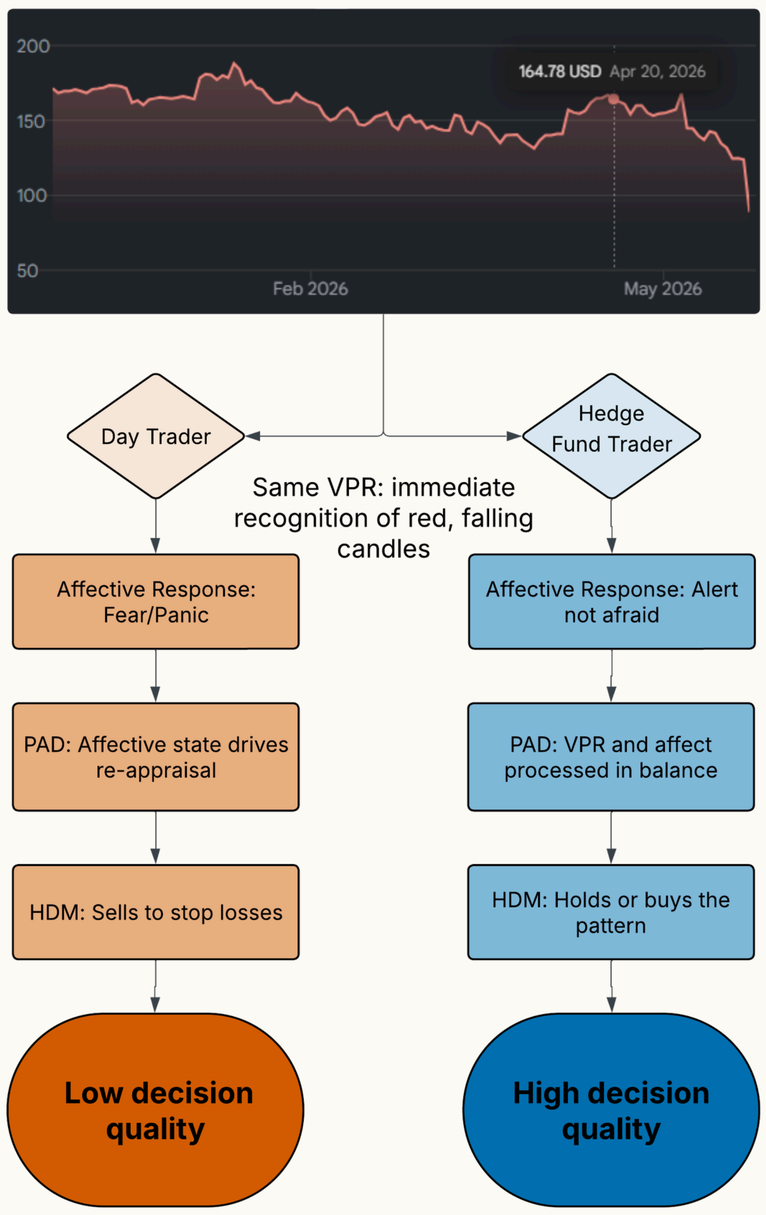
OBJECTIVE

The Visual Perceptual Response-Affect Framework for Integrated Affect and Human Decision-Making, VP-RAFIAH, aims to fill the formal model gap by defining four core components alongside three variables within a visualization's context and modeling how they work together from an HCI perspective.

METHODOLOGY

VP-RAFIAH was developed through an iterative conceptual modeling process. A systematic review across four databases returned 3,138 records, mostly machine-learning work in HCI (CNNs, RNNs, and deep neural networks), with little at the intersection of visualization, affect, and judgment or decision-making. A broader survey across HCI, visualization, cognitive psychology, and decision science followed and returned 29,690 records. After screening, 68 intersected with visualization, affect, and judgment or decision-making. Most were written by researchers in other fields and examined visualizations in their own applications, suggesting a gap in the visualization literature.

WORKING EXAMPLE



COMPONENTS

- VISUAL PERCEPTUAL RESPONSE (VPR)
- PERCEPTUAL-AFFECTIVE DYNAMICS (PAD)
- AFFECT
- HUMAN DECISION-MAKING

VARIABLES

- SCENARIO CONTEXT
- ACTOR
- TASK

LEGEND

- Components are in BLUE rectangles, with lighter to darker indicating the stage in the process.
- Variables are in GOLD diamonds.
- The start and end are in GREEN Ovals.

NEXT STEPS

Our next steps include more evaluation of the framework across domains, standardizing decision quality measurement in empirical contexts, and expanding the framework to multi-actor scenarios where group dynamics and interpersonal affect may influence the vpR-HDM pipeline.

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